# A Robotic RCM-Based Needle Insertion System for Autonomous Abdominal Tumor Resection


## ABSTRACT

This disclosure introduces an improved Remote Center of Motion (RCM) mechanism for surgical devices, such as needle insertion. The design uses a spatial or planar mechanism to maintain the surgical tool's orientation toward the target area. The RCM operates with a single degree of freedom, based on a Parallelogram mechanism.
The modular design allows easy interchange of components and tools, featuring a gear-based lock/unlock system for free needle movement when needed. This streamlined mechanism enhances precision, flexibility, and usability in surgical applications.


## Background of Invention

**[0001]** 1. Field of invention:

**[0002]** Robotics has revolutionized surgical procedures by enhancing precision, minimizing invasiveness, and improving patient outcomes. Robotic systems allow surgeons to perform complex procedures with submillimeter accuracy, reducing tissue damage and optimizing surgical results [1]. They enable smaller incisions, leading to less pain, reduced scarring, and faster recovery times for patients.

**[0003]** Current robotic surgical technology faces several challenges, including high costs, mechanical failures, and steep learning curves. One significant issue is the complexity of performing tasks that require multiple tools to operate synchronously. Many procedures demand the coordination of several robotic arms or even multiple people to handle different instruments simultaneously, which can increase the risk of errors and inefficiencies. For example, in laparoscopic surgeries, surgeons often need to manipulate tools while controlling a camera or gripper, tasks that typically require additional personnel or multiple robotic systems. While advancements like shared control and multi-limb manipulation have enabled single surgeons to control up to four robotic arms using haptic feedback systems, these setups can still be mentally and physically taxing. The need for seamless coordination between robotic arms and intuitive control mechanisms remains a critical area for improvement in robotic surgery.

**[0004]** To address these challenges, there is a need for a simplified yet highly effective mechanism that can simultaneously maintain a surgical tool's alignment with the scanner's field of view, and the target area, while allowing controlled movement. Such a system should minimize complexity, enhance modularity for easy tool interchangeability, and incorporate features to enable free movement when required. Thus there is a need for this invention which can perform the aforesaid.

**[0005]** 2. Prior arts

**[0006]** The field of robotic surgery has experienced transformative advancements in end-effector design. Recent breakthroughs in haptic feedback have addressed one of the most persistent limitations in robotic surgery: the loss of tactile sensation. Systems like Intuitive Surgical's da Vinci 5 now incorporate advanced force-sensing technology that translates tissue resistance into quantifiable feedback for surgeons. [2]. The integration of hybrid haptic models—combining kinesthetic (force-based) and tactile (texture-based) feedback—has further refined precision. For instance, the Senhance Surgical System employs piezoelectric actuators to simulate surface roughness during dissection, reducing unintended tissue damage in laparoscopic procedures [3].

**[0007]** Modularity has emerged as a critical design principle, allowing robotic systems to adapt to diverse surgical tasks. The Versius Surgical Robot by CMR Surgical features a library of quick-swap end effectors, including

bipolar graspers, needle drivers, and ultrasonic scalpels, which can be interchanged mid-procedure without recalibration [4].

**[0008]** Advancements in microfabrication have enabled end effectors capable of operating in sub-millimeter anatomical spaces. The Handheld Robotic Device for Endoscopic Neurosurgery features 3-mm articulated graspers and ring curettes [5]

**[0009]** 3. Ethical and Regulatory Considerations

**[0010]** As end effectors gain autonomy, ethical challenges around liability and safety have intensified. Regulatory bodies like the FDA now require "explainable AI" frameworks for autonomous systems, ensuring surgeons retain override authority[6][7]. Additionally, ISO 13482:2025 mandates fail-safe mechanisms in modular end effectors to prevent accidental detachment during procedures [8]

DESCRIPTION OF INVENTION

**[0011]** The proposed RCM mechanism operates with a single degree of freedom using a Parallelogram-based design. This innovative approach ensures consistent orientation of the surgical tool towards the area of interest while simplifying operation. Additionally, the modular design allows for easy replacement of components and tools, while the gear-based lock/unlock feature provides flexibility for unrestricted needle movement when necessary.
This invention addresses critical gaps in current surgical device technology by offering a compact, precise, and adaptable solution tailored for modern medical applications.

**[0012]** A fixed point in space, where a linkage of a mechanism pivots around. This point may or may not lie on the mechanism. This is an extended concept of Instantaneous center of motion, but the IC stays fixed in space.
This motion is used in Minimally Invasive Surgery(**MIS**), manufacturing etc.

Generally, there are two approaches to obtain RCM output.

1. Serial mechanism with high DOFs and ensure that the end effector mechanism makes the item pivot around a fixed point. But this requires complex control strategies and can cause problems if those fail.
2. A mechanically constrained mechanism whose RCM output is mechanically guaranteed. The Da Vinci robot is a typical example. Parallelograms and pantographs have been implemented in multiple designs.

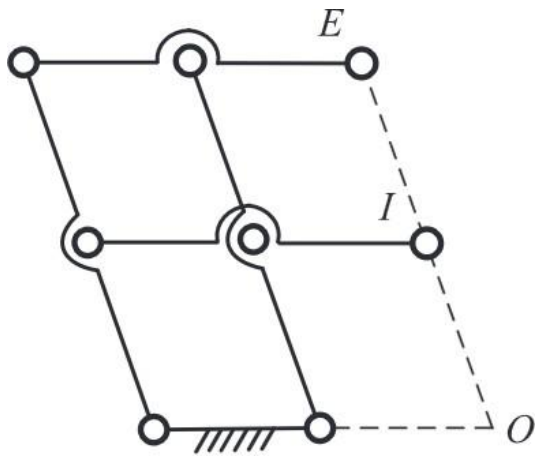



Fig 1. Structure of a parallelogram mechanism

**[0013]** Basic structure of a parallelogram mechanism is shown in Fig 1. It is obvious that the virtual link denoted by **EI** always passes through point **O**. Hence this link gives RCM output.

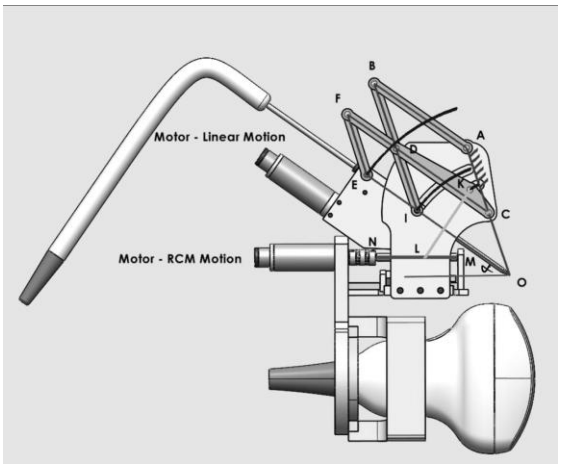



Fig 2. Kinematic linkages of the proposed design

**[0014]** The design proposed uses a double parallelogram and the needle is passed through the virtual link **EI**, and the RCM lies on **O**. The link **AC** is fixed. The assembly to control the linear motion is moved by the attachment at **E** and slot to move the joint **I**.This allows the needle to pivot around RCM point **O**, while the linear motion can be controlled by the motor. The parallelogram is actuated by link **KL** connected to link **KCF**. The link **KL** is attached to a thread linear actuator assembly run by a motor, which moves **L** along the axis **MN**. As **L** moves towards **M** from the position in **Fig 2**., the assembly attains the form in **Fig 3**.

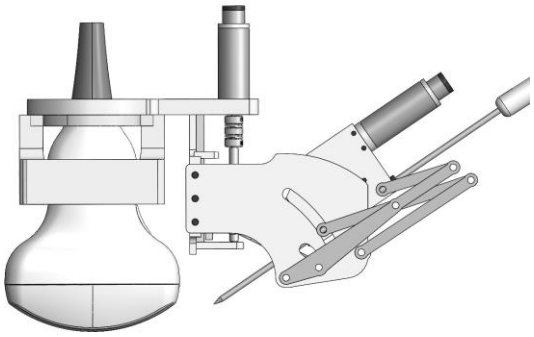


Fig 3. Linkages at the opposite extreme position

**[0015]** According to the embodiment of the present invention, the workspace can be clearly defined. The workspace observed is primarily due to the area swept by needle end. This can be found out by numerical methods, or by visualizing the area swept by keeping any one degree of freedom (needle extension, or swing angle) at an extremity, and moving the other.

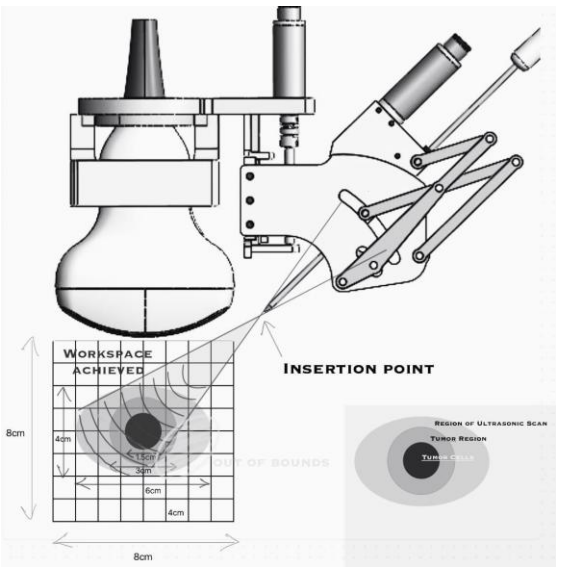



Fig 4. Workspace Achieved by Mechanism

**[0016]** The diagram shows the highlighted workspace area, As it can be seen, the needle sufficiently covers the tumor area, and allows working in a decent range within the viewing region of the Ultrasound Device. A small region outside the Tumor region is inaccessible by this mechanism, but it is still well within the desired region.

DERIVATIONS FOR RELATIONSHIP BETWEEN ANGLE OF NEEDLE WITH RESPECT TO LINEAR POSITION ON LEAD SCREW

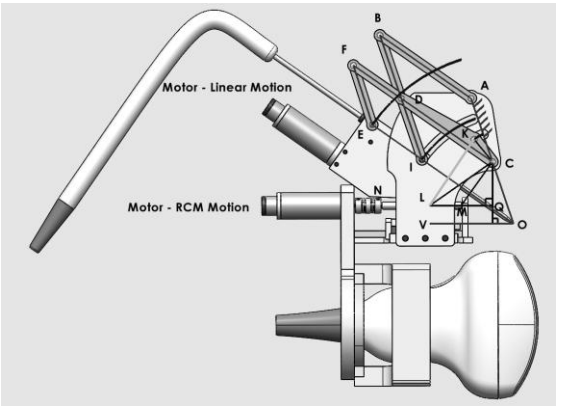



Fig 5. Linkage diagram for derivation

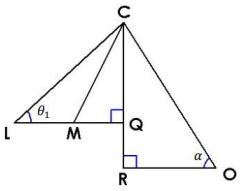
**[0017]** Defining of constant and variable parameters of the mechanism:
The position of **L** can be varied with the rotation of the motor of the lead screw, which will change the angle of the needle (**OI**) from

the vertical. Note that triangles in figure 6-(a), 6-(b), 6-(c), are derived from the same in figure 5.

Input length: - $l_4$, Output angle: - $\alpha$

Mechanism Constraints (Does not change in operation)

| | |
|---|---|
| AB | $l_1$ |
| CK | $l_2$ |
| KL | $l_3$ |
| CQ | $l_5$ |
| MQ | $l_6$ |
| $\angle KCD$ | $\beta$ |




Consider OR to be the vertical w.r.t the mechanism.

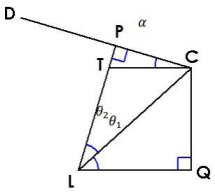
From $\Delta CRO$ and $\Delta CQL$

$CL\cos\theta_1 = ML + MQ$

$\theta_1 = tan^{-1}\left(\frac{CQ}{ML+MQ}\right)$ ***Equation 1.***

$CL = \frac{ML+MQ}{\cos\theta_1}$ ***Equation 2.***

Fig 6(a). Derived $\Delta CRO$ and $\Delta CQL$




From $\Delta PCT, \Delta TCL$ and $\Delta CQL$

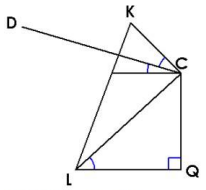
$\angle PCL = 90^0 - \theta_2$

$\angle QCL = 90^0 - \theta_1$

$(90^0 - \theta_2) - \alpha + (90^0 - \theta_1) = 90^0$

$\theta_1 + \theta_2 + \alpha = 90^0$ ***Equation 3.***

Fig 6(b). Derived $\Delta PCT, \Delta TCL$ and $\Delta CQL$




Let $\angle KCL = \theta_3$

$\theta_3 = \beta + (90^0 - \theta_2)$ ***Equation 4.***

$KL^2 = KC^2 + CL^2 - 2.KC.KL.cos\theta_3$

$\theta_3 = cos^{-1}\left(\frac{KC^2+CL^2-KL^2}{2.KC.KL}\right)$ ***Equation 5.***

Fig 6(c). Derived $\Delta KCL$

Calculating $\alpha$ from LM

1. Find $\boldsymbol{\theta_1}$ using **LM** as the input using Eqn 1.
2. Find **CL** using Eqn 2.
3. Find $\boldsymbol{\theta_3}$ using Eqn 5.
4. Find $\boldsymbol{\theta_2}$ using Eqn 4.
5. Find $\boldsymbol{\alpha}$ using Eqn 3.

## Relation Diagrams

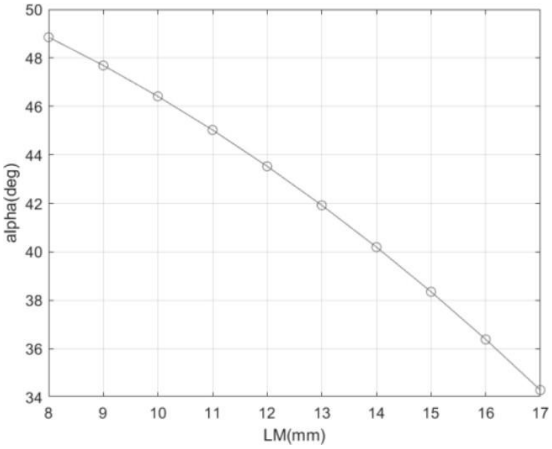
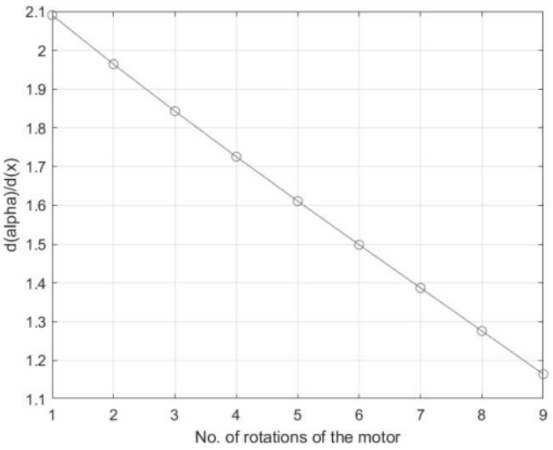



Fig 7(a). Graph: Inclination vs LM




7(b). Graph :$d\alpha/dLM$ vs No. of rotations of the motor (Pitch = 1mm)

**[0018]** According to the embodiment of the present invention, the system incorporates a mechanism for manually overriding automatic needle insertion. This feature enables the surgeon to safely and precisely push or pull the needle to the desired target area through the insertion point located at the Remote Center of Motion (RCM). Such functionality is critical for ensuring modularity and enhancing safety during procedures. It facilitates easy replacement of the needle tip and provides direct force feedback to the surgeon, all while preserving the inherent benefits of the RCM mechanism, such as precision and stability.

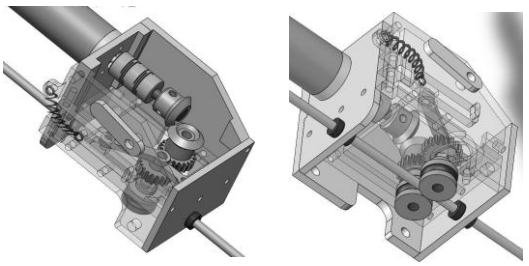
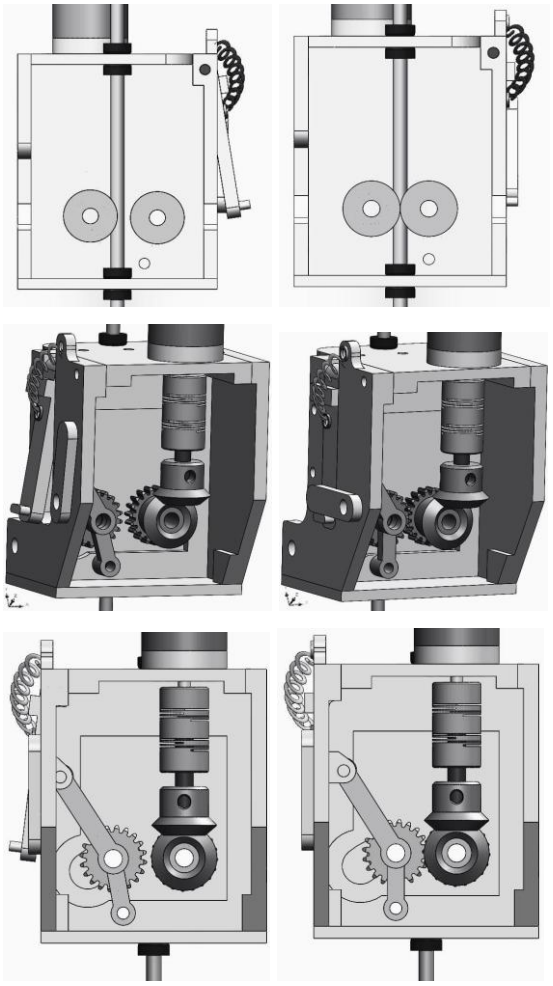


Fig 5. View of Lever Mechanism

**[0019]** This functionality is achieved through a lever mechanism that controls the engagement (figure **8-b**) or disengagement (figure **8-a**) of friction rollers, which are responsible for transmitting motor torque to generate force on the needle. When disengaged, a tensioned spring pulls the lever back to its default position, ensuring accidental engagement is prevented. Conversely, when engagement is required, a

latch mechanism securely locks the lever in place. **In the disengaged mode**, the needle can be manually pushed or pulled by hand, allowing surgeons to apply force directly and receive real-time tactile feedback regarding forces and textures encountered during insertion. This design not only enhances safety and modularity but also ensures precise control while maintaining the advantages of the Remote Center of Motion (RCM) mechanism. **In the engaged mode one** can rely on the precision of the actuators, to insert the needle until a specific depth is reached.



(a) (b)

Fig 8. (a) rollers disengaged. (b) rollers engaged.

**[0020]** Required Components (Non CNC)

| **SCREWS** | |
|---|---|
| m1.6x5 | 20 |
| m2x5 | 12 |
| m2x8 | 6 |
| m3x6 | 4 |
| m3x10 | 9 |
| m3x15 | 1 |
| m3x25 | 2 |
| m4x12 | 3 |
| **OTHERS** | |
| Misumi C-E673ZZ | 2 |
| Misumi SH-CPLS8-2- | 2 |
| Misumi 2-6-2.5 bearing | 1 |
| Misumi 3-6-2.5 bearing | 1 |
| Misumi M50B20_1103_ | 1 |
| Misumi GEAHB0.5-20-3-A-3_b | 1 |
| GEAHB0.5-20-3-A-3_b | 2 |
| Maxon DCX 14L Motor | 2 |

LINKS

RCM Assembly Video
Needle Insertion Assembly Video

RCM Assembly Document
Needle Insertion Assembly Document

Derivations

REFERENCES

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